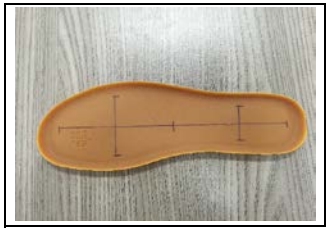
STANDARD OPERATING PROCEDURE OF Shrinkage VF-E13

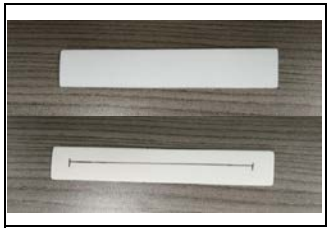
Lab /SOP/ 31



1. Air-circulating oven, capable of maintaining a temperature of 60 ± 1oC and 70 ± 1oC
2. Steel Scale, capable of measuring to 0.5mm.
3. Vernier caliper



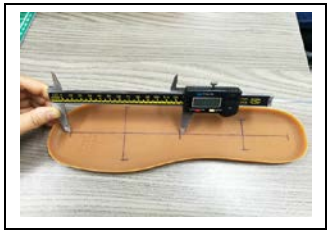
Draw a 5-10mm line from point A to point B. Line AB should be located 1-2cm from the front edge of midsole or unit-sole.
Draw a 5-10mm line from point C to Point D. line CD should be located 1-2cm from the back edge of the midsole or unit-sole.
Draw a line from point E to point F, Line EF should intersect both line AB and line CD.



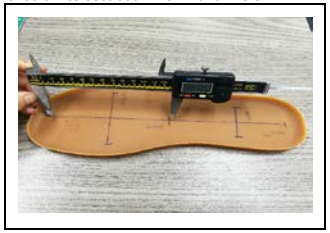
For Sock-liners, Die Cut EVA , Foam , Test plaque Die or hand cut samples which are at least 12.0cm long and have a width of 2.54cm
Draw a line (L) which is 10.0cm long along each of the test samples .line L should be drawn one cm from each edge of the test sample



Check oven calibration, oven temperature must be :
For Expanded components such as MD / IM / Sock- liners, expanded unit-soles or sponge rubber test temperature at 60.0 ± 1.0oC.
For Solid unit soles and outsoles - test temperature at 70.0 ± 1.0oC.



Use a caliper or steel scale,measure the length (L1)
Place the test sample in an oven at 60.0±1.0 0C or 70.0±1.0 0C for 60+/-1 minutes.
Remove the test sample from the oven, and allow it to cool for at least 30 minutes at 23.0+/-2. 0 0C.



Using a Vernier caliper or steel scale, re-measure the length (L2) to the nearest 0.5mm in the same manner.
Shrinkage (%) = L1-L2/L1 *100

Prepared by	Approved by	REVISION		DATE	